

FEDERAL UNIVERSITY OF TECHNOLOGY OWERRI
SCHOOL OF INFORMATION AND COMMUNICATION TECHNOLOGY
DEPARTMENT OF SOFTWARE ENGINEERING

Vital Spring Medical Center

Clinic Appointment System

SYSTEM SPECIFICATION DOCUMENT

Project charter • feasibility study • requirements analysis • data flow diagrams • entity-relationship model • normalization • interface prototypes • database schema • system architecture • implementation and review plan

INS 204 - System Analysis and Design

Group 9 • 200 Level

2025/2026 Academic Session

Project Team - Group 9

Group 9 acted as the systems analysis and design team responsible for preparing the proposed Vital Spring Medical Center Clinic Appointment System.

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Executive Summary

Vital Spring Medical Center is treated in this project as a medium-sized Nigerian outpatient facility that currently relies on paper folders, handwritten registers and a physical appointment book. The proposed Clinic Appointment System replaces the fragmented process with a secure browser-based service for patient registration, practitioner availability, appointment booking, patient check-in, live queue coordination, visit documentation, reminders and management reporting.

The design is distinguished by explicit patient-flow management. A queue entry records when a patient arrives, is called and completes the visit, allowing staff to see delays while they can still act. Patients may book through a self-service channel or receive assistance at reception or by telephone. A daily downtime pack supports continuity during connectivity interruptions.

The report follows the five assignment stages: project charter and feasibility; requirements and data-flow modelling; entity-relationship modelling and normalization; interface, database and architecture design; and implementation review. Preliminary estimates indicate an initial cost of **₦7.30 million**, recurring annual cost of **₦1.40 million**, annual tangible benefits of **₦4.50 million**, and a simple payback period of approximately **2.35 years**. The proposal is technically, economically and operationally feasible under the assumptions stated.

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Project Charter and Feasibility Study

1.1 Project Charter

Item	Definition
Project title	Vital Spring Medical Center Clinic Appointment System
Project sponsor	Vital Spring Medical Center Management
Prepared by	Group 9, INS 204 - System Analysis and Design
Case basis	A local medical center using paper folders and physical appointment books, personalized as Vital Spring Medical Center.

1.1.1 Problem Statement

The assumed existing system uses paper folders and handwritten appointment records. This creates scheduling conflicts, slow retrieval, damaged or misplaced history, repeated data entry, long waits, inconsistent reminders and weak management reporting. Physical records also provide limited control over who views information and no reliable record of access.

- **Conflicting appointments:** the same practitioner and time may be allocated twice.
- **Slow patient identification:** variations in names and phone numbers create duplicate folders.
- **Unmanaged waiting:** staff know that patients are present but cannot see a consistent queue or waiting duration.
- **Lost continuity:** unavailable history can delay care and cause repeated questions.
- **Administrative burden:** staff copy information and count registers manually.
- **Missed visits:** reminders depend on individual telephone calls.
- **Limited evidence:** attendance, no-shows, waiting time and practitioner utilization are difficult to measure.

1.1.2 Project Goals

ID	Goal	Measure of Success
G1	Centralize active patient and appointment information.	Authorized search returns a patient in under 5 seconds.
G2	Prevent practitioner scheduling conflicts.	No two active appointments reserve one availability slot.
G3	Support self-service and assisted booking.	Reception, telephone and patient portal bookings use one schedule.
G4	Make patient arrival and waiting visible.	Every checked-in appointment has a queue status and timestamps.
G5	Improve access to relevant patient history.	Authorized practitioners can open previous visits before consultation.
G6	Reduce missed appointments.	Confirmation and reminder outcomes are recorded for valid contacts.
G7	Provide operational reports.	Attendance, wait-time and utilization reports generate by date range.
G8	Protect patient confidentiality.	Role restrictions and activity logs cover sensitive functions.

1.2 Scope

In Scope

- User authentication and role permissions
- Patient registration, search and update
- Departments and practitioner profiles
- Dated availability slots
- Booking, rescheduling and cancellation
- Reception and telephone-assisted booking

Out of Scope

- Medical billing and payments
- Insurance and HMO claims
- Pharmacy inventory
- Laboratory instrument integration
- Inpatient admission and bed allocation
- Payroll and staff recruitment

- Patient check-in and live queue tracking
- Relevant visit history and new visit records
- Confirmations and reminders
- Attendance, wait-time and utilization reports
- Activity logging and backups
- Daily downtime appointment pack

- Ambulance dispatch
- Video consultation
- External hospital record exchange

1.3 Key Stakeholders

Stakeholder	Interest	Project Involvement
Clinic Manager	Service performance, cost and accountability	Approves scope and reviews reports
Doctors	Accurate schedules, queues and history	Interviews, prototype review and acceptance tests
Nurses	Arrival order and permitted patient information	Workflow validation and training
Receptionists	Fast registration and conflict-free booking	Primary workflow participants
Records Staff	Patient identity quality and migration	Defines matching and verifies converted records
Patients	Accessible booking, shorter waits and privacy	Usability feedback and self-service use
System Administrator	Accounts, availability, backup and support	Technical review and operation

1.4 Technical Feasibility

The required technology is established and available in Nigeria. A browser interface, server application and relational database avoid specialized clinic hardware while supporting central rules and backups.

Component	Requirement	Assessment
Client devices	Reception, records, clinical and management computers; modern browsers	Common hardware; phased replacement possible
Hosting	Managed server or professionally maintained local server	Available; managed hosting reduces local administration
Database	Relational DBMS with transactions and constraints	Mature technology suited to appointments
Network	Secured router, local network and primary internet link	Available; downtime pack mitigates interruptions
Power	UPS and backup electricity for essential workstations	Required due to local power risk
Messaging	SMS or email service with delivery response	Available from commercial providers
Backup	Encrypted daily copy stored separately and restoration tests	Technically straightforward but must be monitored

Technical conclusion

The project is technically feasible. Its principal dependencies are protected workstations, power continuity, reliable hosting, competent support and tested recovery.

1.5 Economic Feasibility and Simple Cost-Benefit Analysis

All amounts are classroom estimates in Nigerian naira, not quotations. Actual procurement requires current prices and verified clinic baselines.

Initial Cost Item	Estimate (₦)
System analysis and design	450,000
Application development and testing	2,200,000
Computers and peripherals	1,600,000

Network equipment and installation	400,000
Power backup equipment	600,000
Initial hosting and security setup	250,000
Active-record conversion	350,000
Staff training	300,000
Initial maintenance and support	500,000
Contingency allowance	650,000
Total Initial Cost	7,300,000

Annual Operating Item	₦	Annual Tangible Benefit	₦
Hosting, backup and security	360,000	Administrative time saved	1,500,000
Maintenance and support	600,000	Recovered appointment capacity	2,000,000
Messages and connectivity	350,000	Paper and storage reduction	450,000
Refresher training	90,000	Reduced correction effort	300,000
		Faster reporting	250,000
Total	1,400,000	Total	4,500,000

Simple calculation

Annual net benefit = ₦4,500,000 - ₦1,400,000 = **₦3,100,000**.

Payback period = ₦7,300,000 ÷ ₦3,100,000 = **2.35 years**, approximately two years and four months.

Intangible benefits include better patient confidence, continuity of care, staff coordination and accountability. The project is economically feasible if the planning assumptions are confirmed.

1.6 Operational Feasibility

- **Workflow fit:** registration, booking, check-in and consultation remain familiar, but are coordinated through one record.
- **Training:** reception and records personnel receive scenario-based training; clinical users focus on queue and visit documentation; administrators practise recovery.
- **Accessibility:** assisted booking remains available for patients without internet access or digital confidence.
- **Transition:** a controlled pilot and temporary parallel checks reduce disruption.
- **Privacy:** individual accounts, role restrictions, screen locking, minimal reminder content and confidentiality training apply.
- **Downtime:** the next-day appointment list and numbered sheets allow safe temporary operation, followed by verified back-capture.

Operational conclusion

The system is operationally feasible when management supports the change, users receive practical training, active records are verified, and both digital and downtime procedures are rehearsed.

Chapter 1 establishes that the defined appointment and patient-flow system is feasible and should proceed to detailed requirements analysis.

Requirements Analysis and Process Modelling

Requirements represent what stakeholders need from the proposed system and the qualities that make those functions acceptable. Interviews, observation of registration and appointment handling, examination of paper forms and short patient questionnaires are proposed as gathering methods. The requirements below remain traceable to the process and data designs.

2.1 User Stories

ID	As a...	I want...	Acceptance Criteria
US-01	Patient	to see available dates and request an appointment without visiting the center.	Only open slots appear; a successful booking returns a reference and confirmation.
US-02	Patient	to change or cancel an appointment so an unusable slot is released.	The current booking is identified; permitted changes update the schedule and send a notice.
US-03	Patient	to receive a reminder so I can prepare and arrive on time.	A reminder is queued before confirmed appointments and the delivery result is recorded.
US-04	Receptionist	to find or register a patient and complete assisted booking quickly.	Search accepts number, phone or name; likely duplicates are shown before creation.
US-05	Receptionist	to check in an arriving patient and issue a queue number.	A valid appointment creates one queue entry with check-in time and waiting status.
US-06	Nurse	to see the live queue so patients can be prepared in order.	Waiting, vitals, called and completed states update without re-entering patient details.
US-07	Doctor	to publish consultation slots so bookings reflect my actual availability.	Slots contain date and time; blocked or reserved slots cannot be offered.
US-08	Doctor	to view relevant visit history and document the current visit.	Only authorized clinical users can open history; the record links to the appointment.
US-09	Manager	to view attendance, waiting-time and utilization reports.	Reports filter by period and department and totals reconcile with appointment statuses.
US-10	Administrator	to manage accounts and inspect sensitive activity.	Accounts can be activated or disabled; defined access and change events identify the user and time.

2.2 Functional Requirements

Ref	Area	The System Shall...
FR-01	Authentication	authenticate every staff user before protected functions are available.
FR-02	Authorization	enforce role permissions for receptionist, records, nurse, doctor, manager and administrator functions.
FR-03	Patient	create a patient profile with a unique hospital number and required identity and contact information.
FR-04	Patient	search by hospital number, phone or name and warn about likely duplicate patients.
FR-05	Practitioner	maintain departments, practitioner profiles and active status.
FR-06	Availability	create, block and display dated practitioner availability slots.
FR-07	Appointment	book one open slot for one patient through portal, front desk, telephone or walk-in source.
FR-08	Appointment	support confirmation, permitted rescheduling and cancellation while preserving history.
FR-09	Conflict control	recheck and reserve a slot atomically so two appointments cannot use it.
FR-10	Queue	check in a valid appointment and create one numbered queue entry.
FR-11	Queue	track waiting, vitals, called, with-practitioner, completed and left statuses with timestamps.
FR-12	Visit record	allow authorized practitioners to view relevant history and document complaint, findings, diagnosis and care plan.

FR-13	Notification	send confirmation, reminder, reschedule and cancellation messages and record delivery outcomes.
FR-14	Reporting	generate attendance, no-show, cancellation, waiting-time and practitioner-utilization reports.
FR-15	Administration	create, disable and manage staff accounts without exposing password values.
FR-16	Activity	log defined login, search, view and data-change actions with user and timestamp.
FR-17	Continuity	produce a next-day appointment list and accept verified back-capture after downtime.
FR-18	Backup	support daily protected backups and authorized restoration.

2.3 Non-Functional Requirements

Ref	Quality	Requirement
NFR-01	Privacy	Patient and health information shall be available only to authenticated roles with a work-related need.
NFR-02	Security	Network traffic shall use HTTPS; passwords shall be salted hashes; staff sessions shall expire after 15 minutes of inactivity.
NFR-03	Performance	Routine search and booking screens shall respond within 5 seconds for 25 concurrent users under normal connectivity.
NFR-04	Availability	The service target shall be 99% monthly availability excluding approved maintenance.
NFR-05	Recovery	The recovery time target shall be 4 hours and maximum planned data loss 24 hours.
NFR-06	Usability	A trained receptionist shall locate an existing patient and book an open slot within 2 minutes.
NFR-07	Accessibility	Forms shall have labels, keyboard operation, readable contrast and errors not communicated by color alone.
NFR-08	Integrity	Keys, references, valid status values and one-appointment-per-slot rules shall be enforced in the database.
NFR-09	Auditability	Activity entries shall identify the user, action, object and time and shall not be editable by ordinary application users.
NFR-10	Maintainability	Components, database rules, backup steps and account procedures shall be documented.
NFR-11	Scalability	The design shall support three times the assumed starting patient volume without changing the core data model.
NFR-12	Message privacy	Patient messages shall omit diagnosis and detailed visit reasons.

2.4 Context-Level Data Flow Diagram

The context diagram treats the complete Vital Spring system as one process. It defines the boundary between internal processing and five external entities. Every flow is named as data rather than an action.

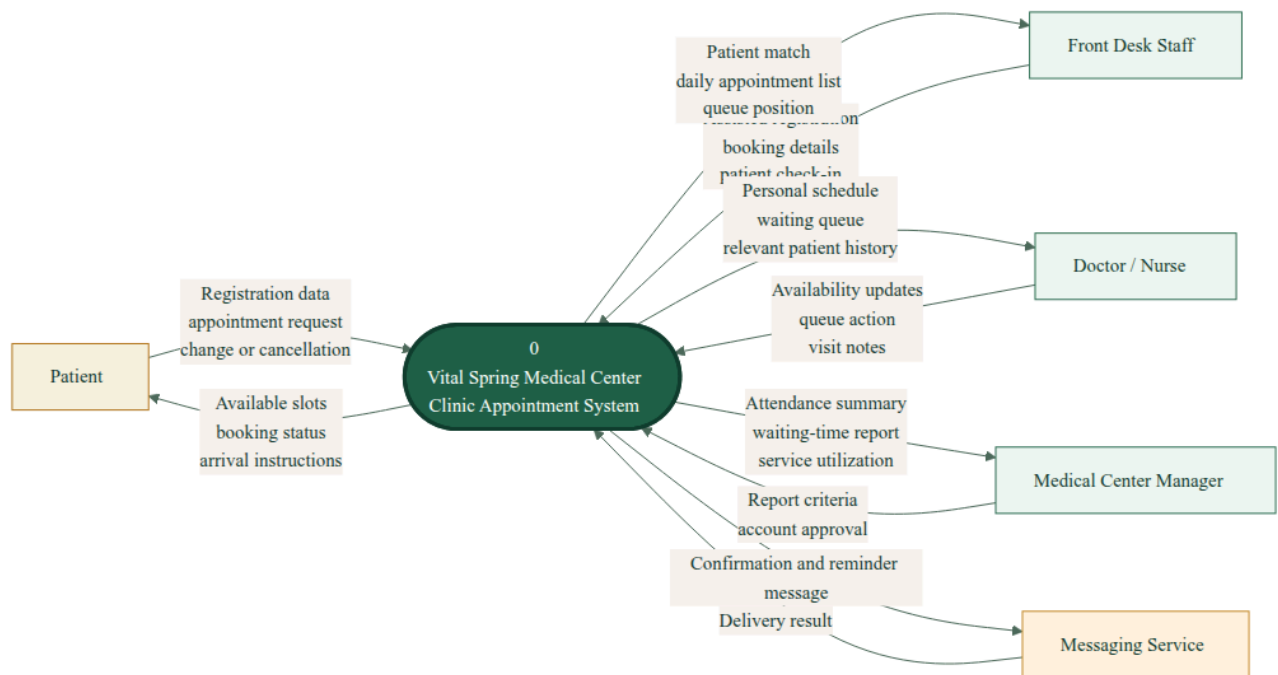


Figure 2.1: Context-Level Data Flow Diagram for the Vital Spring Medical Center Clinic Appointment System.

Patients exchange registration, booking and change information. Front desk staff support registration, appointments and check-in. Doctors and nurses provide schedules, queue actions and visit notes. Management requests reports and account approvals. The messaging service is external because it carries notifications but does not own clinic records.

2.5 Level-0 Data Flow Diagram

The Level-0 DFD decomposes the system into six processes and six logical data stores. Its distinctive patient-flow process separates arrival management from appointment coordination, allowing waiting-time information to be captured without treating an appointment and an arrival as the same event.

Data Modelling and Normalization

3.1 Entities and Attributes

Entity	Purpose	Principal Attributes
PATIENT	Identifies a person receiving care.	patient_id (PK), hospital_number, name, DOB, sex, phone, email, address, emergency contact
DEPARTMENT	Groups practitioners by service area.	department_id (PK), department_name, location
PRACTITIONER	Represents a doctor or other bookable professional.	practitioner_id (PK), department_id (FK), staff_number, full_name, role, contacts, active
AVAILABILITY_SLOT	A dated period offered by one practitioner.	slot_id (PK), practitioner_id (FK), slot_date, start_time, end_time, slot_status
APPOINTMENT	Reserves one slot for one patient.	appointment_id (PK), patient_id, practitioner_id, slot_id, recorded_by (FKs), source, reason, status
QUEUE_ENTRY	Tracks the patient's physical arrival and progress.	queue_entry_id (PK), appointment_id (FK), check-in time, queue number, status, called/completed times
VISIT_RECORD	Stores the outcome of a completed consultation.	visit_record_id (PK), appointment_id (FK), complaint, findings, diagnosis, care plan, time
NOTIFICATION	Records appointment messages and delivery results.	notification_id (PK), appointment_id (FK), type, channel, destination, status, time
USER_ACCOUNT	Identifies an authorized staff user and role.	user_id (PK), username, password_hash, role, display_name, active, last_login
ACTIVITY_LOG	Records defined sensitive system actions.	activity_id (PK), user_id (FK), action, object, object_id, time, source_address

3.2 Relationships, Cardinality and Optionality

Relationship	Cardinality	Business Rule
DEPARTMENT contains PRACTITIONER	1 : 0..many	Every practitioner belongs to one department; a new department may have none.
PRACTITIONER offers AVAILABILITY_SLOT	1 : 0..many	Every slot belongs to one practitioner.
PATIENT requests APPOINTMENT	1 : 0..many	Every appointment names one patient.
PRACTITIONER receives APPOINTMENT	1 : 0..many	Every appointment names one practitioner.
AVAILABILITY_SLOT is reserved by APPOINTMENT	1 : 0..1	An open or blocked slot has no appointment; a reserved slot has at most one.
APPOINTMENT creates QUEUE_ENTRY	1 : 0..1	Cancelled and no-show appointments may have no arrival entry.
APPOINTMENT results in VISIT_RECORD	1 : 0..1	A completed visit may produce one signed record.
APPOINTMENT generates NOTIFICATION	1 : 0..many	Confirmation, reminder, reschedule and cancellation are separate messages.
USER_ACCOUNT records APPOINTMENT	1 : 0..many	recorded_by is optional for patient self-service bookings.
USER_ACCOUNT performs ACTIVITY_LOG	1 : 0..many	Each staff activity entry identifies one account.

3.3 Entity-Relationship Diagram

The crow's-foot ERD expresses the logical data structure. Queue entries and availability slots are deliberately independent entities: an appointment reserves a planned slot, while the queue records what happened after physical arrival.

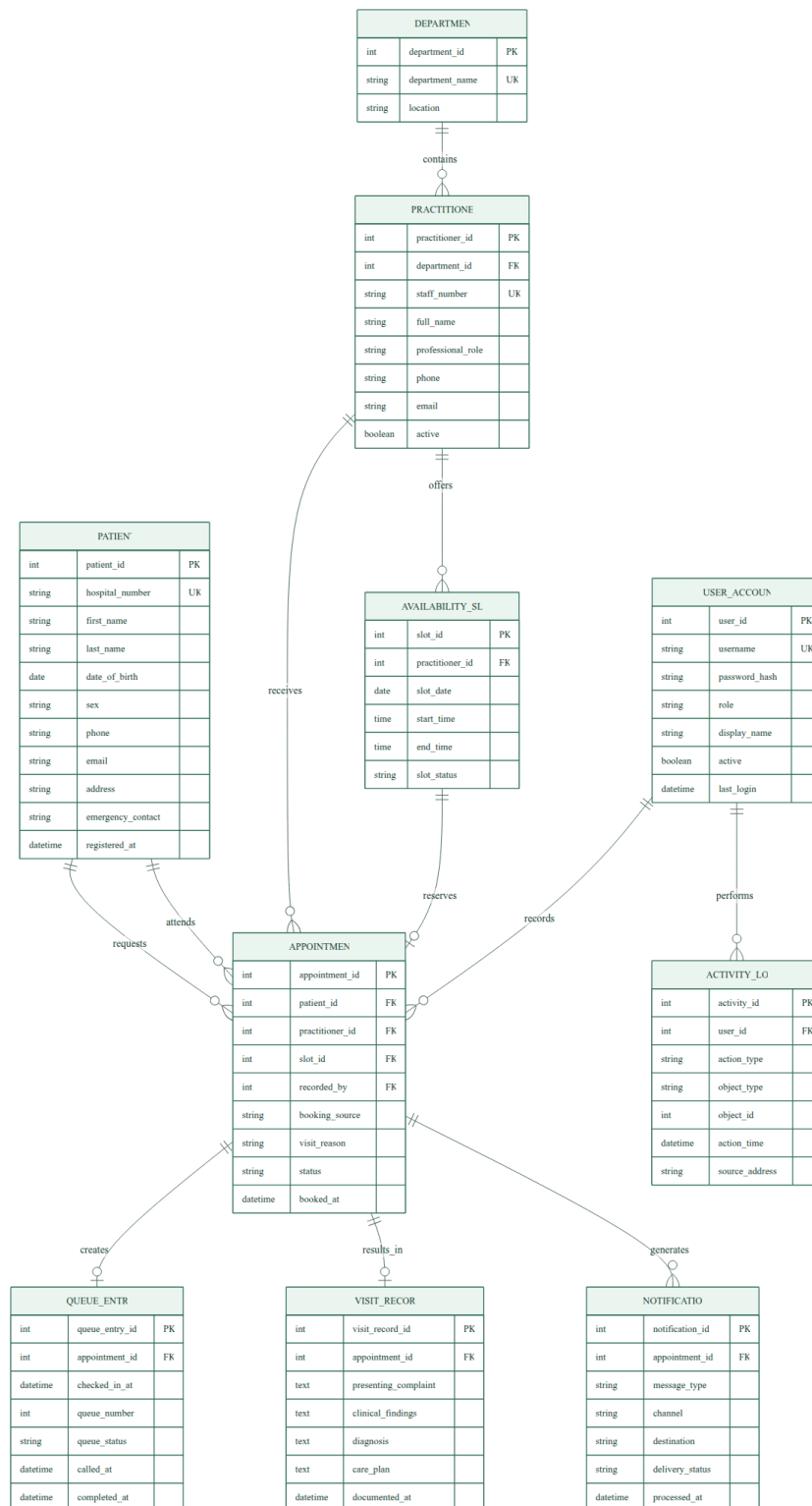


Figure 3.1: Entity-Relationship Diagram with keys, attributes, cardinality and optionality.

3.4 Normalization to Third Normal Form

Normalization begins with a plausible paper register containing patient details, practitioner details and multiple visits in one row. It then removes repeating groups, partial dependencies and transitive dependencies.

3.4.1 Unnormalized Form (UNF)

Hospital No.	Patient Name	Phone	Appointments {Date, Time, Practitioner, Department, Location, Queue No., Outcome}
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VS-002184	Amarachi Okafor	08031244017	{04/08/26, 09:00, Dr Nkiru Ibekwe, General Outpatient, Block A, A08, Completed}; {18/08/26, 10:20, Dr Nkiru Ibekwe, General Outpatient, Block A, -, Booked}
VS-001637	Suleiman Bala	07055881220	{04/08/26, 11:20, Dr Adaora Madu, Physiotherapy, Annex 2, A12, Waiting}

The appointment group repeats inside a single cell. The number of columns is not fixed, values are not atomic, and date-based appointment searches require reading inside each patient row.

3.4.2 First Normal Form (1NF)

Each appointment becomes one row and every field contains one value. A temporary composite key is (HospitalNo, SlotDate, StartTime).

HospitalNo	PatientName	Phone	SlotDate	Time	PractitionerNo	PractitionerName	Department	Location	QueueNo
VS-002184	Amarachi Okafor	08031244017	04/08/26	09:00	PR-014	Dr Nkiru Ibekwe	General Outpatient	Block A	A08
VS-002184	Amarachi Okafor	08031244017	18/08/26	10:20	PR-014	Dr Nkiru Ibekwe	General Outpatient	Block A	-
VS-001637	Suleiman Bala	07055881220	04/08/26	11:20	PR-021	Dr Adaora Madu	Physiotherapy	Annex 2	A12

The table is atomic but patient facts repeat per appointment. Practitioner and department facts also repeat, producing update, insertion and deletion anomalies.

3.4.3 Second Normal Form (2NF)

PatientName and Phone depend only on HospitalNo, which is part of the temporary composite key. They are moved to PATIENT. Appointment data receives an AppointmentID. Practitioner information still remains repeated in APPOINTMENT at this stage.

PATIENT			APPOINTMENT_2NF					
HospitalNo (PK)	Name	Phone	ID	HospitalNo	Date/Time	PractitionerNo	Name	Department
VS-002184	Amarachi Okafor	08031244017	A-801	VS-002184	04/08 09:00	PR-014	N. Ibekwe	General Outpatient
VS-001637	Suleiman Bala	07055881220	A-802	VS-002184	18/08 10:20	PR-014	N. Ibekwe	General Outpatient

3.4.4 Third Normal Form (3NF)

PractitionerName and Department depend on PractitionerNo rather than AppointmentID. DepartmentLocation depends on DepartmentID. Queue information depends on the patient's arrival, not merely on the appointment. These transitive and separate-event facts are extracted.

3NF Relation	Attributes
PATIENT	PatientID (PK), HospitalNumber, Name, DOB, Sex, Phone, Email, Address, EmergencyContact
DEPARTMENT	DepartmentID (PK), DepartmentName, Location
PRACTITIONER	PractitionerID (PK), DepartmentID (FK), StaffNumber, FullName, Role, Contacts, Active
AVAILABILITY_SLOT	SlotID (PK), PractitionerID (FK), SlotDate, StartTime, EndTime, SlotStatus
APPOINTMENT	AppointmentID (PK), PatientID, PractitionerID, SlotID, RecordedBy (FKs), Source, Reason, Status, BookedAt
QUEUE_ENTRY	QueueEntryID (PK), AppointmentID (FK), CheckedInAt, QueueNumber, QueueStatus, CalledAt, CompletedAt
VISIT_RECORD	VisitRecordID (PK), AppointmentID (FK), Complaint, Findings, Diagnosis, CarePlan, DocumentedAt

Normalization Summary

UNF to 1NF: removed repeating appointment groups. **1NF to 2NF:** separated patient facts that depended on only part of the temporary key. **2NF to 3NF:** separated practitioner, department and queue facts that depended on other non-key facts or distinct events. The final relations store each fact in one appropriate place.

The normalized relations correspond directly to the ERD and form the basis of the physical schema in Chapter 4.

System Design and Interface Prototyping

4.1 Design Principles

- **Patient flow first:** the dashboard prioritizes who is present, how long each patient has waited and what should happen next.
- **Progressive tasks:** registration and booking use visible steps rather than one crowded form.
- **Error prevention:** search precedes registration, unavailable slots cannot be selected, and the server rechecks every reservation.
- **Assisted access:** reception staff can perform the same core booking task for patients who cannot use self-service.
- **Minimum necessary information:** administrative lists omit diagnosis and detailed clinical notes.
- **Accessible feedback:** text labels accompany color, controls have clear names, and invalid fields retain correct input.

4.2 Live Patient-Flow Dashboard

The dashboard is not a conventional list of today's appointments. Its central element is the arrival queue, supported by waiting-time indicators, next-appointment context and a delay alert. This lets clinic staff respond before waiting becomes excessive.

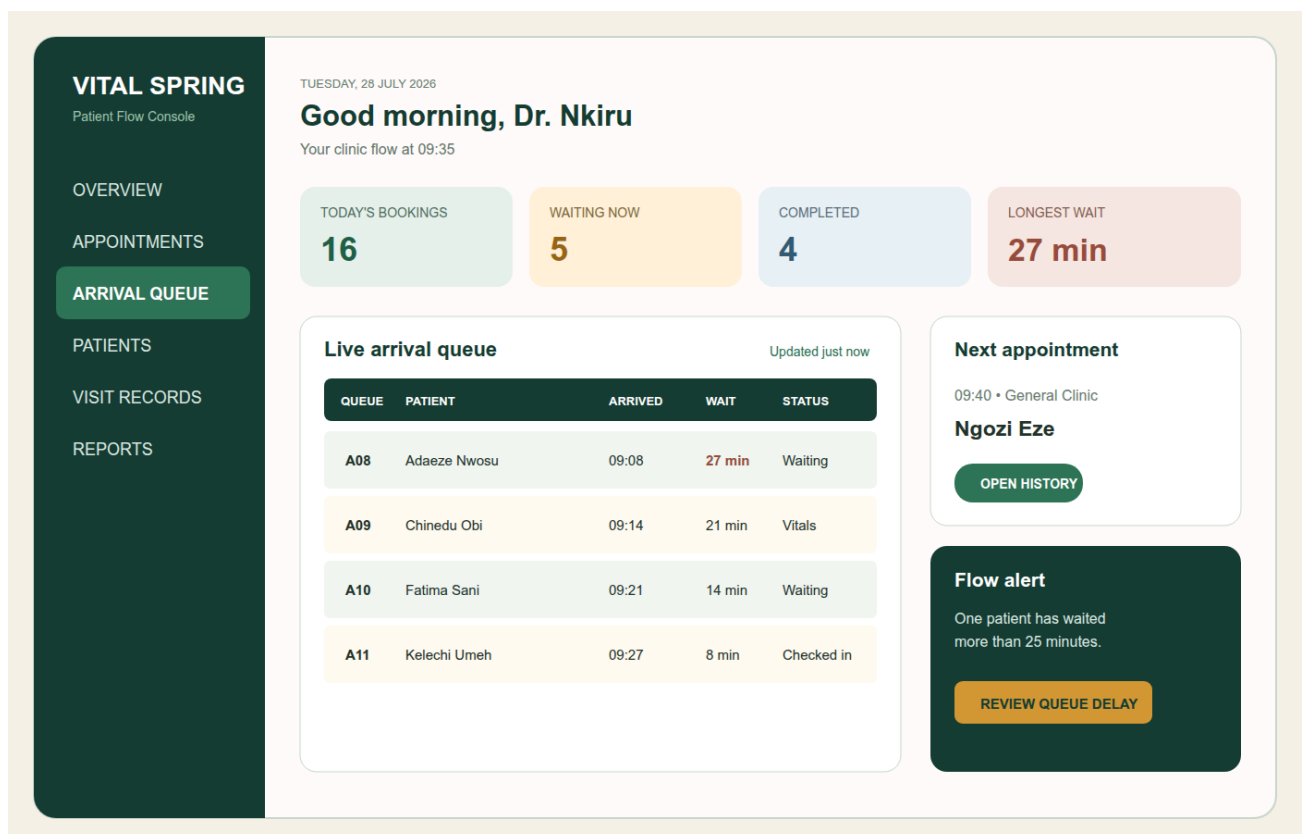


Figure 4.1: Doctor and staff dashboard centered on live arrivals and waiting time.

The status column remains understandable in grayscale because each state is written in text. The history action is available only when the signed-in role has clinical permission.

4.3 Patient Registration Form

The registration flow starts with an existing-patient search. Identity, contact, emergency and review stages reduce cognitive load. The example displays a phone error without clearing the valid name and date fields.

Vital Spring Medical Center
Register a patient

1 Identity
2 Contact
3 Emergency
4 Review

Patient identity

Search before creating a new record to avoid duplicates.

FIRST NAME *

LAST NAME *

DATE OF BIRTH *

SEX *

PHONE NUMBER *

EMAIL ADDRESS

Enter an 11-digit Nigerian phone number.

☐ I have explained how Vital Spring uses and protects patient information. *

CANCEL

CONTINUE

Figure 4.2: Staged patient registration with duplicate search, inline validation and privacy acknowledgement.

Field	Required	Validation
First and last name	Yes	2-50 characters; letters, spaces, apostrophes and hyphens.
Date of birth	Yes	Valid date not later than today; review warning for implausible age.
Sex	Yes	Selected from approved values.
Phone	Yes	Normalized Nigerian number; duplicate search reruns after correction.
Email	No	Valid address format when supplied.
Address	Yes	5-160 characters.
Emergency contact	Yes	Name and reachable phone.
Privacy acknowledgement	Yes	Staff confirms explanation before final registration.

4.4 Appointment Booking Form

The booking flow selects department and practitioner before a date. Only open, future slots appear. Booked slots remain visible but unavailable so users understand why a desired time cannot be selected.

Book an appointment

Assisted booking • Patient VS-002184

1. Select service and practitioner

DEPARTMENT

General Outpatient Clinic

PRACTITIONER

Dr. Nkiru Ibekwe

2. Choose a date

MON
03

TUE
04

WED
05

THU
06

FRI
07

3. Select an available slot

08:20

08:40 booked

09:00

09:20

10:20

10:40

11:00 booked

11:20

REASON FOR VISIT *

Persistent lower back pain

Booking summary

PATIENT
Amarachi Okafor

SERVICE
General Outpatient Clinic

PRACTITIONER
Dr. Nkiru Ibekwe

DATE AND TIME
Tuesday 04 August
09:00 – 09:20

CONFIRM BOOKING

The slot will be checked again before confirmation.

Figure 4.3: Assisted appointment booking with dated slots and a persistent confirmation summary.

At confirmation, the system locks the selected slot, checks that it is still open, creates the appointment and marks the slot reserved in one transaction. If another user reserved it first, the transaction does not create a duplicate booking and the user receives newly available alternatives.

4.5 Attendance and Patient-Flow Report

The management report combines conventional attendance outcomes with median waiting time and practitioner utilization. The flow insight draws attention to a period that requires staffing or scheduling review.

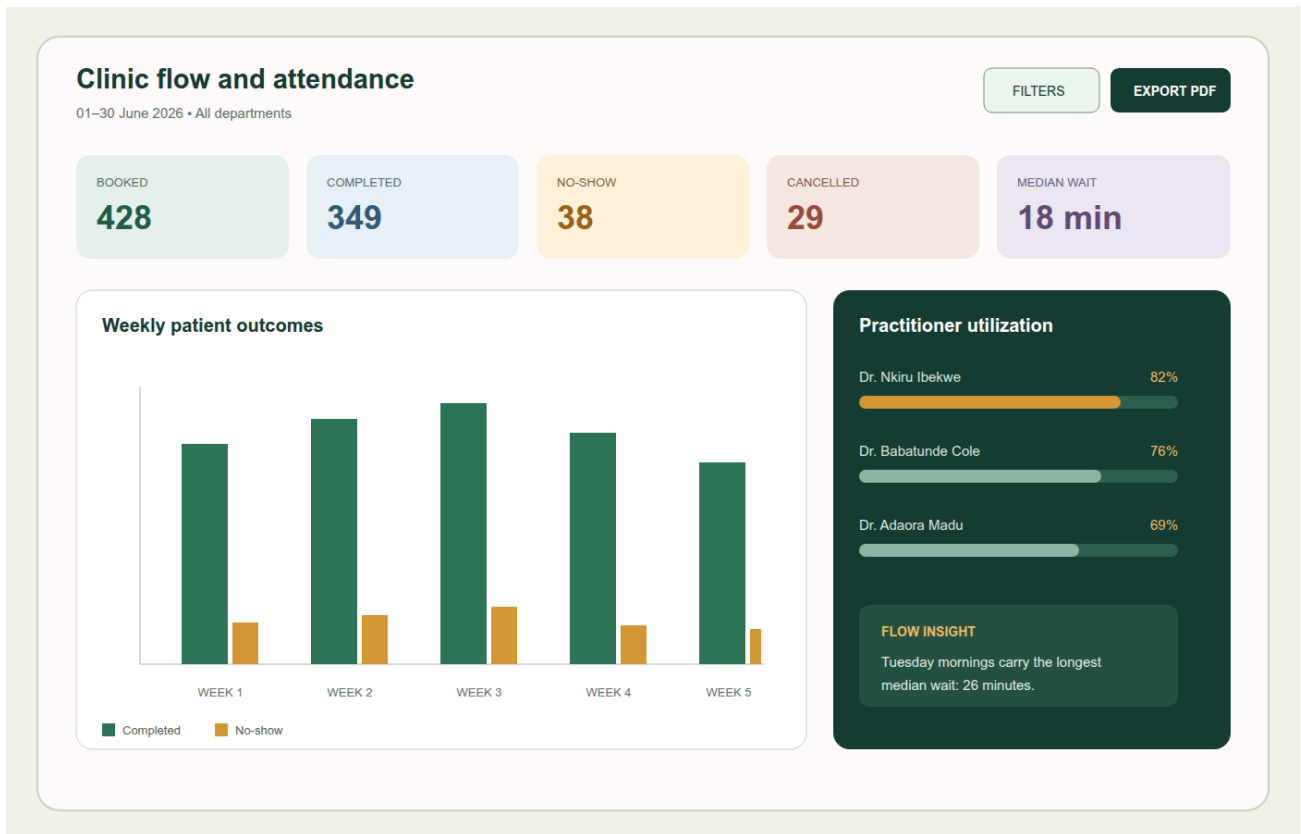


Figure 4.4: Attendance, no-show, waiting-time and practitioner-utilization report.

All displayed numbers are illustrative project data. Production values would be calculated from appointment outcomes, queue timestamps and practitioner slots for the selected reporting period.

4.6 Physical Database Schema

Table	Important Columns and Constraints
department	department_id BIGSERIAL PK; department_name VARCHAR(80) UNIQUE; location VARCHAR(80)
user_account	user_id PK; username UNIQUE; password_hash; role CHECK; display_name; active; last_login
patient	patient_id PK; hospital_number UNIQUE; identity, contact and emergency fields; registered_at
practitioner	practitioner_id PK; department_id FK; staff_number and email UNIQUE; professional_role; active
availability_slot	slot_id PK; practitioner_id FK; date and times; slot_status CHECK; UNIQUE practitioner/date/start
appointment	appointment_id PK; patient, practitioner, slot and recorded_by FKs; slot_id UNIQUE; source and status CHECK
queue_entry	queue_entry_id PK; appointment_id UNIQUE FK; check-in, called and completion timestamps; status CHECK
visit_record	visit_record_id PK; appointment_id UNIQUE FK; complaint, findings, diagnosis and care plan
notification	notification_id PK; appointment_id FK; message type, channel and delivery status CHECK
activity_log	activity_id PK; user_id FK; action, object, time and source address

4.6.1 Appointment Conflict Control

Database and transaction rule

1. Begin a transaction and lock the requested availability slot.
2. Verify that the slot is still open and that its practitioner matches the request.
3. Insert the appointment. The unique slot_id permits only one appointment for that slot.
4. Update the slot to reserved and commit.
5. If any check fails, roll back and return current alternatives.

Cancelled appointments may reopen a future slot according to the cancellation rule. Historical appointment rows remain for reporting and accountability.

4.7 Proposed System Architecture

Vital Spring uses a secure client-server architecture with four access channels. All channels pass through one protected web gateway; none connects directly to the database. Application services separate identity, scheduling and queue, visit documentation, reporting and background messaging.

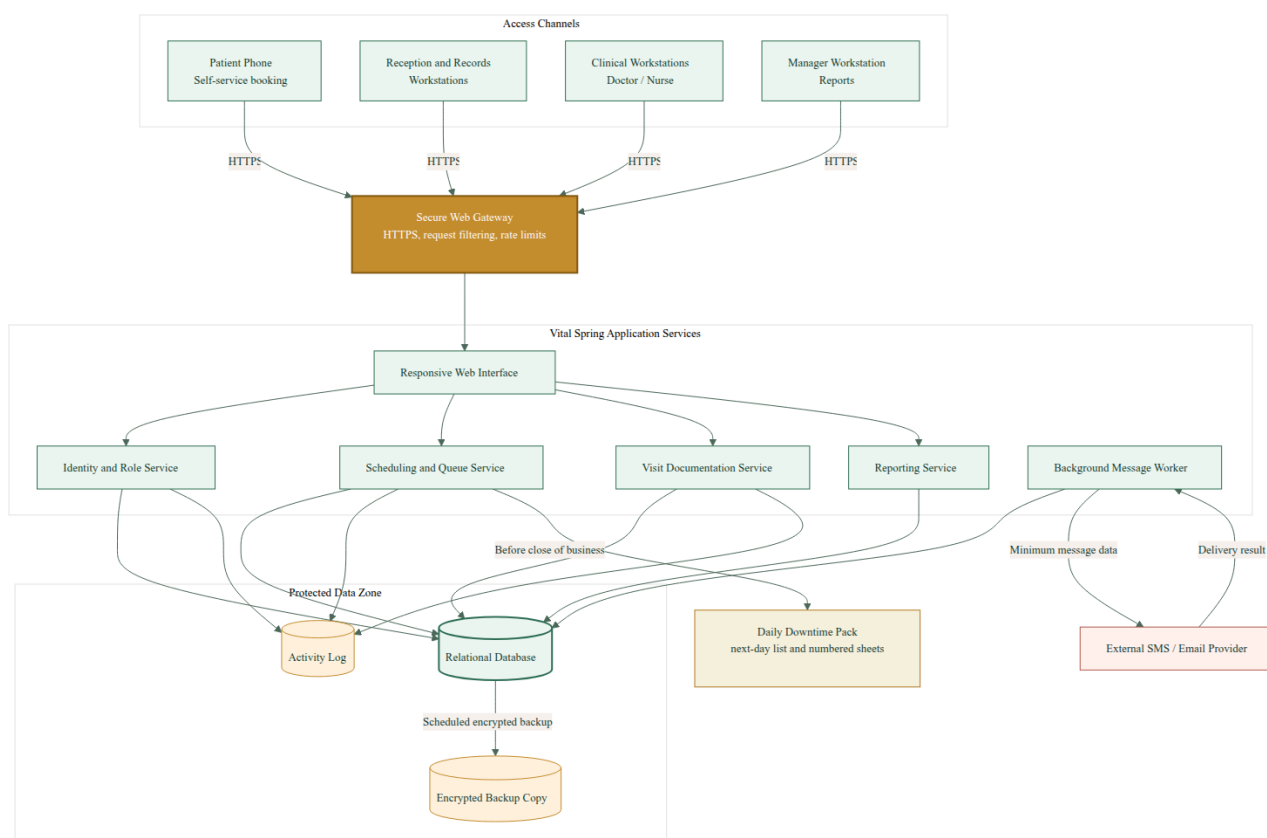


Figure 4.5: Proposed architecture including protected data services, external messaging and the daily downtime pack.

Component	Responsibility
Access channels	Role-specific patient, front desk, clinical and management interfaces.
Secure web gateway	HTTPS termination, request filtering and traffic limits.
Identity and role service	Authentication, session handling and role checks.
Scheduling and queue service	Slot reservation, appointment lifecycle, check-in and queue timestamps.
Visit documentation service	Permitted history retrieval and current visit records.
Reporting service	Attendance, waiting-time and utilization calculations.
Background message worker	Queued messages, retries and provider delivery results.
Relational database	Transactional operational data with keys and constraints.
Activity log	Evidence of defined sensitive actions.
Encrypted backup	Separate recovery copy.
Daily downtime pack	Minimum printed continuity information for the next clinic day.

4.8 Security and Patient Privacy

Function	Reception	Records	Nurse	Doctor	Manager	Administrator
Patient identity	Create/edit	Create/edit	View	View	Limited	Support only
Appointments	Create/edit	View	View/check-in	Own schedule	View/report	Support
Queue	Check-in	No	Manage	Call/complete	Report	Support
Visit records	No	No clinical content	Limited view	View/write for care	No clinical content	No routine content
Reports	No	No	No	Own summary	Full operational	Technical

User accounts	No	No	No	No	Approval	Manage
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- Unique accounts, strong password hashes and multi-factor authentication for privileged roles.
- Server-side permission checks and 15-minute inactive-session timeout.
- HTTPS for all clients and encryption for database disks and backups where supported.
- Minimum reminder content with no diagnosis or detailed visit reason.
- Activity logging for login, patient search, record access, export and data changes.
- Daily backups, separate credentials and quarterly restoration exercises.
- Retention, correction and incident procedures aligned with the Nigeria Data Protection Act 2023 and professional confidentiality.

Chapter 4 turns the logical models into interfaces, database rules and a deployable architecture while preserving patient-flow and privacy requirements.

Implementation and Review Plan

5.1 Phased Implementation

Phase	Weeks	Activities and Exit Condition
1. Confirm	1-2	Observe workflows, validate requirements and measure assumed baselines. Exit: approved requirements.
2. Prototype	3-4	Test registration, booking, queue and report screens with representative users. Exit: agreed revisions.
3. Build	5-12	Implement identity, patient, practitioner, slot, appointment, queue, visit, message and report functions. Exit: integrated test release.
4. Test	13-16	Functional, integrity, security, performance, usability and recovery tests. Exit: no unresolved critical defect.
5. Prepare Data	14-18	Inventory, clean, capture and verify active paper records. Exit: signed reconciliation.
6. Train	17-18	Role-based scenarios and privacy training. Exit: competence demonstrated.
7. Pilot	19-20	One service area uses the system with daily issue review and controlled comparison. Exit: stable core workflow.
8. Deploy	21	Final migration, account activation and on-site support. Exit: authorized operational use.
9. Review	22, 30 and 90 days	Compare metrics, incidents, user feedback and costs. Exit: acceptance or corrective plan.

Downtime Procedure

Reception prints the next working day's appointment list before closing. During an outage, staff use numbered sheets containing only necessary identity and appointment fields. After restoration, one staff member enters the activity and another verifies it. Clinical paper notes are signed and transferred only by an authorized clinician.

5.2 Paper-Record Migration

1. Inventory active folders and assign each a migration tracking number.
2. Identify likely duplicates using hospital number, name, date of birth and phone.
3. Define which identity, contact, appointment-summary and clinically necessary fields will be captured.
4. Enter information into a restricted staging area; do not use personal devices.
5. Have a second person verify identity and critical clinical fields against the folder.
6. Reconcile counts and investigate every rejected or duplicate row.
7. Import approved records in controlled batches and record the result.
8. Retain paper originals for the approved validation period; dispose only through authorized confidential handling.

5.3 Testing Strategy

Test Level	Representative Evidence
Unit	Phone normalization, slot status, queue duration, appointment transitions and report formulas.
Integration	Foreign keys, slot transaction, message provider responses and activity logging.
System	Every FR tested through valid, invalid and boundary scenarios.
Security	Role bypass, unauthorized record access, injection, session expiry, account disabling and export restrictions.
Performance	25 concurrent users; common screens meet 5-second target.
Recovery	Restore a protected backup into an isolated environment within four hours.
Usability	Representative users perform their core tasks without coaching.

User Acceptance	Manager, receptionist, nurse and doctor sign agreed fictional scenarios.
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5.3.1 Selected Acceptance Tests

ID	Scenario	Expected Result
AT-01	Two users confirm the same Slot 09:00 simultaneously.	Exactly one appointment commits; the other receives alternatives.
AT-02	Reception user attempts to open a visit-record address.	Access denied; no clinical text returned; denied attempt logged.
AT-03	Registration uses an existing phone and matching DOB/name.	Possible duplicate displayed before any new patient is created.
AT-04	Reception checks in one appointment twice.	One queue entry remains; second attempt explains current queue number.
AT-05	Message provider reports temporary failure.	Notification becomes retrying and later records final delivery result.
AT-06	Backup restoration exercise is performed.	Service starts in isolation within four hours and sample records reconcile.
AT-07	Manager filters June General Outpatient report.	Totals equal underlying appointment states; median wait uses queue timestamps.

5.4 Training Plan

Audience	Duration	Content	Competence Evidence
All staff	1 hour	Confidentiality, credentials, screen privacy, phishing and incident reporting	80% privacy assessment before activation
Reception and records	2 half-days	Search, duplicates, registration, slots, appointment changes, check-in and downtime	Complete five timed scenarios
Nurses	Half-day	Arrival validation, queue states, permitted information and escalation	Process one arrival-to-call scenario
Doctors	Half-day	Slots, personal queue, history, visit records and corrections	Complete a fictional visit
Manager	2 hours	Reports, utilization, account approval and activity review	Generate and explain a report
Administrator	1 day	Accounts, monitoring, backups, restoration, updates and incidents	Complete restoration and account lifecycle exercises

5.5 Post-Implementation Review

All baselines below are project assumptions until measured during Phase 1. The 90-day review compares like-for-like clinic periods.

Metric	Assumed Baseline	90-Day Target	Source
Double-booked slots	Occurs manually	0	Slot and appointment exception report
Patient search time	5-15 minutes	Under 5 seconds	Timed sample
Existing-patient booking time	8-12 minutes	2 minutes or less	Workflow observation
No-show rate	To be measured	15% relative reduction	Appointment statuses
Median arrival-to-call wait	To be measured	20% reduction	Queue timestamps
Assisted/self-service mix	100% manual	Monitor, no exclusion target	Booking source
Availability	Not applicable	99% monthly	Monitoring report
Backup restoration	Untested	Quarterly test within 4 hours	Signed exercise report
User task completion	Not measured	90% without coaching	Training and 30-day review

Overall Recommendation

Proceed with the Vital Spring Medical Center Clinic Appointment System within the defined outpatient scope. The design addresses registration, scheduling, arrivals, visit documentation, reminders and reporting while preserving assisted access, downtime continuity and patient confidentiality.

The implementation plan converts the specification into controlled phases with measurable acceptance, training and operational review.

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Scope of Statements

This student specification provides a proposed system design. Financial and operational quantities are project assumptions, and the privacy discussion does not replace professional legal or security review before real patient data is processed.

Appendix A - Requirements Traceability Matrix

Requirement	Design Evidence	Verification
FR-01, FR-02; NFR-01, 02	User Account, Activity Log, Process 6.0, role matrix, identity service	Login, timeout, disabled-account and role-bypass tests
FR-03, FR-04	Process 1.0, D1, PATIENT, registration mockup	Valid/invalid registration and duplicate AT-03
FR-05, FR-06	Process 2.0, D2, DEPARTMENT, PRACTITIONER, AVAILABILITY_SLOT	Slot boundary and blocked-slot tests
FR-07-09; NFR-08	Process 3.0, booking mockup, slot transaction and unique slot_id	Booking source, reschedule and AT-01
FR-10, FR-11	Process 4.0, D4, QUEUE_ENTRY, dashboard	AT-04 and timestamp/status tests
FR-12	Process 5.0, D5, VISIT_RECORD, role matrix	Visit save, history and unauthorized AT-02
FR-13; NFR-12	Process 6.0, NOTIFICATION, message worker	Message content, schedule, retry AT-05
FR-14	Report mockup, reporting service, Appointment and Queue data	Formula reconciliation and AT-07
FR-15, FR-16	USER_ACCOUNT, ACTIVITY_LOG, administration design	Account lifecycle and append-only permission tests
FR-17	Downtime pack and implementation procedure	Simulated outage and two-person back-capture
FR-18; NFR-05	Encrypted backup component	AT-06 restoration exercise
NFR-03, 04, 06, 07	Architecture and responsive mockups	Load, timing, keyboard and contrast tests

Appendix B - Glossary

Term	Meaning
Activity log	A protected record identifying who performed a defined system action and when.
Cardinality	The number of instances of one entity that may relate to another.
Data Flow Diagram (DFD)	A model of external entities, processes, data stores and data moving between them.
Entity-Relationship Diagram (ERD)	A model of entities, attributes, identifiers and relationships.
Foreign Key (FK)	An attribute that references the primary key of another relation.
Normalization	Organizing relations to reduce repetition and data anomalies.
Primary Key (PK)	An attribute or combination that uniquely identifies a row.
Queue Entry	A record of a patient's physical arrival and progress through the outpatient queue.
Role-Based Access Control	Permissions assigned according to defined job responsibilities.
Recovery Point Objective	The maximum planned period of data that may need to be recreated after failure.
Recovery Time Objective	The target time for restoring an unavailable service.
User Acceptance Testing	Testing by representative users against agreed business requirements.

Vital Spring Medical Center

Care in motion. Information in place.

End of System Specification Document